[Total: 12]

5 Cycling is a demanding endurance sport that pushes athletes to optimise every aspect of their performance. Chemists play a critical role in this field by enhancing bicycle materials, improving energy metabolism in cyclists and in ensuring safety.

(a) Cyclists often look for ways to reduce the weight of their bicycles, which typically weigh around 7.4 kg. One proposed idea is to inflate bicycle tyres with helium instead of air to reduce weight.

(i) State two basic assumptions of kinetic theory as applied to an ideal gas.

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.....

..... **[2]**

(ii) Using the data in Table 5.1, calculate the respective mass of helium and mass of air required under the same given conditions.

Suggest, with a reason, whether the use of helium provides a significant advantage in terms of mass.

Table 5.1

	Value
Molar mass of helium (He)	4.0 g mol^{-1}
Molar mass of air (approximate)	29.0 g mol^{-1}
Volume of gas in a standard bicycle tyre	2.0 dm^3
Pressure in tyre	8 bar
Temperature	298 K

.....

..... **[3]**

(iii) Rubber tyres are made of vulcanised rubber, a cross-linked polymer. Although they appear solid, they contain tiny free volumes between polymer chains at the nanometer scale, typically around 0.3 – 0.5 nm.

With reference to the *Data Booklet*, suggest why helium should not be used to inflate the tyres.

……

…… **[1]**

(b) A bicycle frame must balance tensile strength, weight, durability and cost. Choosing the right material is key to optimising cycling performance.

Tensile strength is the maximum stress that a material can withstand before it shows significant deformation of its body shape.

Table 5.2

Material	Density (g/cm^3)	Tensile Strength (MPa)	Relative Cost	Corrosion Resistance
Aluminium	2.70	310	Moderate	Moderate
Titanium	4.50	900	High	High
Steel	7.85	500	Low	Low
Graphite Fibre	1.60	600	Very High	High

(i) Explain, in terms of structure and bonding, why graphite fibre has relatively high tensile strength.

.....

..... **[1]**

(ii) Considering the data provided in Table 5.2, recommend the most suitable material for a high-performance racing bicycle frame. Justify your choice in terms of the factors in Table 5.2.

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..... **[2]**

Commented [SXF3]: Delete comma

Commented [SXF4R3]: I feel that the qns there is far too much data. So can the 4 marks to allocate to others questions to test syllabus stuff?

Commented [KCH5R3]: To be discussed.

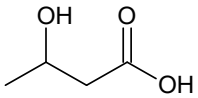
Commented [LW(6]: Include definition of tensile strength

(c) Endurance athletes, such as cyclists, are constantly seeking ways to boost stamina and fight fatigue. One option is to consume BHB energy supplement.

Beta-hydroxybutyrate (BHB), is a lab-made compound that serves as an efficient fuel source for both the brain and body when glucose levels are low.

In cells, BHB enters the mitochondria to produce ATP, which is the body's main source of energy.

Compared to glucose, it generates less waste, helps conserve NAD^+ (a molecule essential for energy metabolism) and avoids blood sugar spikes. However, BHB is also expensive, has a bitter taste, may cause nausea and is absorbed more slowly than glucose. This makes it less ideal for short, intense bursts of energy.



beta-hydroxybutyrate (BHB)

(i) State the systematic name for BHB.

..... **[1]**

(ii) BHB has stereoisomers. State the type of stereoisomerism present in BHB and draw the stereoisomers.

Type of stereoisomerism: ……

[2]

(iii) Use the data in Table 5.3, calculate the energy released in kJ g^{-1} for both BHB and glucose when they undergo combustion.

Table 5.3

	Molar mass / g mol^{-1}	Standard enthalpy change of combustion / kJ mol^{-1}
BHB	118.13	−2430
Glucose	180.16	−2805

[1]

(iv) Based on your calculations in **(c)(iii)**, suggest whether using BHB as an energy supplement would benefit endurance cyclists. Give a reason for your answer.

……

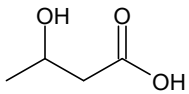
…… **[1]**

(v) Suggest a disadvantage of using BHB as an energy supplement.

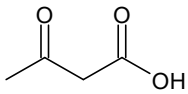
……

…… **[1]**

(d) BHB, acetoacetate (AcAc) and acetone are fat-derived compounds made mainly in the liver. They form about 80% of the ketones in the blood and provide energy especially to the brain during fasting, intense exercise or low-carbohydrate diets.



beta-hydroxybutyrate (BHB)



acetoacetate (AcAc)

acetone

Describe a simple chemical test, with appropriate observations, that can be carried out to distinguish between each of the following pairs of compounds.

- AcAc and acetone
- AcAc and BHB

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..... **[4]**

[Total: 19]

END OF PAPER